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Adaptive Degraded Document Image Binarization

Basilis Gatos, Ioannis Pratikakis, Stavros J. Perantonis
Computational Intelligence Laboratory, Institute of Informatics and Telecommunications
National Center for Scientific Research "Demokritos", GR-153 10 Agia Paraskevi, Athens, Greece
Email: {bgat, ipratika, sper}@iit.demokritos.gr

Abstract—Document image binarization is the most critical preprocessing stage in Optical Character Recognition (OCR) systems. Degraded document images frequently suffer from shadows, non-uniform illumination, low contrast, large signal-dependent noise, smear, and ink bleed-through, causing conventional global and local thresholding techniques to fail. This paper presents a novel, five-stage adaptive digital image preprocessing methodology for binarizing and enhancing severely degraded grayscale document images without requiring user-tuned morphological parameters. The pipeline consists of: (1) a low-pass adaptive Wiener pre-filtering step to suppress background noise while preserving stroke edges; (2) a rough estimation of foreground regions using Niblack's local approach; (3) explicit computation of the background intensity surface by interpolating neighboring background pixels; (4) final adaptive thresholding that combines the estimated background surface with the filtered image while adapting to local text contrast; and (5) post-processing via iterative shrink and swell morphological filtering to eliminate noise speckles and repair broken character contours. Extensive experimental evaluations using quantitative OCR recognition rates and Levenshtein edit distance confirm that the proposed method consistently outperforms state-of-the-art thresholding techniques.

Keywords—Document Image Binarization; Image Preprocessing; Optical Character Recognition (OCR); Adaptive Wiener Filter; Background Surface Estimation; Degraded Documents.

1. Introduction

Binarization of grayscale document images is the foundational first step in document analysis and Optical Character Recognition (OCR) pipelines [1]. Poor illumination in camera-captured documents, aging of historical manuscripts, ink seepage from the reverse side of pages (bleed-through), and stain marks produce complex intensity variations across the document page [2]. Existing binarization techniques are broadly classified into two categories: global thresholding methods and local adaptive methods.

Global thresholding algorithms, most notably Otsu's variance-maximization method [3], compute a single intensity threshold T for the entire image based on bimodal histogram assumptions. While computationally fast, global thresholding fails completely when uneven illumination or shadows cause background intensity in one part of the page to fall below foreground text intensity in another region. Local adaptive thresholding methods—such as Bernsen [4], Niblack [5], and Sauvola & Pietikäinen [6]—compute a spatially varying threshold $T(x, y)$ within a local sliding window of size $W \times W$.

Although Niblack's method successfully detects low-contrast character strokes, it introduces severe ghost noise in text-free background regions. Sauvola's modification suppresses background noise by scaling the local standard deviation by its dynamic range, yet it frequently thins or breaks faint character strokes under heavy degradation [6]. To overcome both limitations simultaneously, we propose a five-stage methodology that explicitly models the background surface and adapts the threshold function to local stroke contrast before applying connectivity-preserving post-processing.

2. Stage 1: Pre-Processing via Adaptive Wiener Filtering

Because degraded documents contain high-frequency sensor noise and uneven paper grain texture, directly applying local thresholding amplifies background fluctuations. Standard linear low-pass filters (such as box mean filters) blur character edges and merge adjacent strokes. Instead, given a source grayscale document image $I_s(x, y) \in [0, 255]$, we apply a 2D pixel-wise adaptive Wiener filter over a local $M \times N$ neighborhood (typically 3×3) [7]. The filtered image $I(x, y)$ is defined as:

$$I(x, y) = \mu(x, y) + [(\sigma^2(x, y) - \nu^2) / \sigma^2(x, y)] \cdot (I_s(x, y) - \mu(x, y)) \quad (1)$$

where $\mu(x, y)$ and $\sigma^2(x, y)$ represent the local mean and local variance of intensity values inside the $M \times N$ window centered at (x, y), and ν^2 denotes the average of all local estimated variances across the entire page:

$$\mu(x, y) = (1 / MN) \sum_{i,j} I_s(i, j), \quad \sigma^2(x, y) = (1 / MN) \sum_{i,j} I_s^2(i, j) - \mu^2(x, y) \quad (2)$$

In uniform background areas where local variance $\sigma^2(x, y) \approx \nu^2$, the Wiener filter performs strong smoothing ($I(x, y) \approx \mu(x, y)$). Near character strokes where $\sigma^2(x, y) \gg \nu^2$, very little smoothing occurs, preserving sharp character boundaries.

3. Stage 2: Rough Estimation of Foreground Regions

The goal of the second stage is to obtain a preliminary binary map $S(x, y) \in \{0, 1\}$ that captures all true foreground text pixels (even faint strokes), allowing some false positives in the background so that no character pixels are inadvertently treated as background during surface interpolation. We apply Niblack's local adaptive thresholding [5] to the Wiener-filtered image $I(x, y)$:

$$T_{Niblack}(x, y) = m(x, y) + k \cdot s(x, y) \quad (3)$$

where $m(x, y)$ and $s(x, y)$ are the sample mean and standard deviation in a sliding window of size $W \times W$ (e.g., 60×60 pixels), and parameter $k = -0.2$. Pixels with $I(x, y) < T_{Niblack}(x, y)$ are labeled as rough foreground candidates $S(x, y) = 1$, while remaining pixels are labeled as background $S(x, y) = 0$.

4. Stage 3: Background Surface Estimation

Next, we construct a continuous estimate of the underlying document background illumination surface $B(x, y)$. For pixels classified as background ($S(x, y) = 0$), the background surface equals the filtered image value directly. For pixels classified as rough foreground ($S(x, y) = 1$), the background intensity is interpolated from surrounding background pixels within a local window of size $d_x \times d_y$ chosen to exceed the maximum character stroke thickness:

$$B(x, y) = I(x, y) \text{ if } S(x, y) = 0; \quad B(x, y) = [\sum I(i, j)(1 - S(i, j))] / [\sum (1 - S(i, j))] \text{ if } S(x, y) = 1 \tag{4}$$

Unlike morphological closing approaches that require manual selection of structuring element geometry [2], this direct spatial interpolation accurately reconstructs non-uniform shadows and page folds without distorting nearby character regions.

5. Stage 4: Final Adaptive Thresholding

Text extraction is achieved by comparing the local intensity difference $B(x, y) - I(x, y)$ against a spatially adaptive distance threshold $d(B(x, y))$. Because dark shadow regions exhibit compressed dynamic range between ink and paper compared to brightly illuminated areas, $d(B(x, y))$ must scale non-linearly with the local background level $B(x, y)$. First, the average stroke contrast δ and average background intensity b over the document are computed as:

$$\delta = [\sum_{x,y} (B(x, y) - I(x, y))] / [\sum_{x,y} S(x, y)], \quad b = [\sum_{x,y} B(x, y)(1 - S(x, y))] / [\sum_{x,y} (1 - S(x, y))] \tag{5}$$

The final binary image $T(x, y) \in \{0, 1\}$ and the sigmoid contrast-adaptive threshold function $d(B(x, y))$ are then formulated as:

$$T(x, y) = 1 \text{ if } B(x, y) - I(x, y) > d(B(x, y)), \text{ else } 0; \quad d(B(x, y)) = q \cdot \delta \cdot [(1 - p_2) / (1 + \exp((-4B(x, y)) / (b(1 - p_1)) + (2(1 + p_1)) / (1 - p_1))) + p_2] \tag{6}$$

where empirically validated parameters $q = 0.6$, $p_1 = 0.5$, and $p_2 = 0.8$ ensure that regions with dark backgrounds use a proportionally smaller threshold distance $q \cdot \delta \cdot p_2$, preserving weak text strokes under deep shadows while eliminating bleed-through and stain artifacts in bright regions.

6. Stage 5: Post-Processing via Shrink and Swell Filtering

To further refine the binary image $T(x, y)$ for OCR outline tracing, an iterative sequence of conditional binary morphological filters—termed Shrink and Swell filters [8]—is applied within a local $n \times n$ neighborhood:

- **Shrink Filter:** Eliminates isolated foreground noise pixels. If a foreground pixel ($T(x, y) = 1$) has fewer than P_{sh} foreground neighbors in its $n \times n$ window, it is reset to background (0).
- **Swell Filter:** Fills small bays, pinholes, and micro-breaks along character strokes. If a background pixel ($T(x, y) = 0$) has more than P_{sw} foreground neighbors inside its $n \times n$ window, it is set to foreground (1) provided it does not bridge two previously disconnected character components.

7. Experimental Evaluation and OCR Recognition Impact

The proposed five-stage method was benchmarked against five widely cited thresholding algorithms: Otsu's global method **[3]**, Bernsen's local contrast method **[4]**, Niblack's method **[5]**, Yanowitz and Bruckstein's adaptive surface method **[9]**, and Sauvola and Pietikäinen's adaptive method **[6]**. Evaluations were performed on degraded historical manuscripts, shadowed camera-captured book pages, and stain/bleed-through documents. End-to-end OCR performance was measured using ABBYY FineReader Engine via character recognition rate and Levenshtein edit distance against ground-truth transcriptions, alongside pixel-level Precision, Recall, and F-Measure.

Table I. Comparative Quantitative Evaluation: Pixel-Level Binarization Metrics and End-to-End OCR Recognition Accuracy

Binarization Method	Category	Pixel Precision (%)	Pixel Recall (%)	F-Measure (%)	Levenshtein Error	OCR Character Accuracy (%)
Otsu (1979) **[3]**	Global Histogram	64.2%	78.5%	70.6%	148.4	61.4%
Bernsen (1986) **[4]**	Local Contrast	68.9%	81.2%	74.5%	126.0	67.2%
Niblack (1986) **[5]**	Local Mean & Std	59.4%	94.1%	72.8%	134.2	65.1%
Yanowitz & Bruckstein (1989) **[9]**	Edge Surface	84.1%	85.6%	84.8%	64.5	83.2%
Sauvola & Pietikäinen (2000) **[6]**	Local Adaptive	89.7%	86.4%	88.0%	44.1	88.5%
Proposed 5-Stage Method (Without Stage 5)	Adaptive Surface	92.8%	93.2%	93.0%	21.8	94.3%
Proposed 5-Stage Method (Full Pipeline)	**Adaptive + Shrink/Swell**	**95.4%**	**94.8%**	**95.1%**	**11.2**	**97.1%**

As shown in Table I, Otsu's global threshold fails on shadowed and stained pages (61.4% OCR accuracy), while Niblack's method achieves high pixel recall (94.1%) but suffers from low precision (59.4%) due to severe background noise that confuses OCR page layout segmentation. Sauvola's method improves precision significantly (89.7%, yielding 88.5% OCR accuracy) but loses faint character strokes in low-contrast regions. By contrast, the proposed five-stage methodology preserves both high stroke recall (94.8%) and high background cleanliness (95.4% precision), achieving a 97.1% OCR character recognition rate and reducing the Levenshtein edit error by nearly 75% compared to Sauvola's method.

8. Conclusions

We have presented a complete, parameter-robust classical image preprocessing pipeline for degraded document binarization and OCR enhancement. By integrating adaptive Wiener noise filtering, Niblack rough foreground detection, explicit background surface interpolation, contrast-adaptive sigmoid thresholding, and connectivity-preserving shrink/swell post-processing, the method successfully handles heavy shadows, uneven illumination, and ink bleed-through, substantially elevating downstream OCR accuracy.

References

1. O. D. Trier and T. Taxt, "Evaluation of binarization methods for document images," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 17, no. 3, pp. 312–315, Mar. 1995.
2. M. Sezgin and B. Sankur, "Survey over image thresholding techniques and quantitative performance evaluation," *Journal of Electronic Imaging*, vol. 13, no. 1, pp. 146–168, Jan. 2004.
3. N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, Jan. 1979.
4. J. Bernsen, "Dynamic thresholding of grey-level images," in *Proc. 8th Int. Conf. Pattern Recognition (ICPR)*, Paris, France, Oct. 1986, pp. 1251–1255.
5. W. Niblack, *An Introduction to Digital Image Processing*. Englewood Cliffs, NJ, USA: Prentice Hall, 1986, pp. 115–116.
6. J. Sauvola and M. Pietikäinen, "Adaptive document image binarization," *Pattern Recognition*, vol. 33, no. 2, pp. 225–236, Feb. 2000.
7. J. S. Lim, *Two-Dimensional Signal and Image Processing*. Englewood Cliffs, NJ, USA: Prentice Hall, 1990, pp. 536–540.
8. R. Schalkoff, *Digital Image Processing and Computer Vision*. New York, NY, USA: John Wiley & Sons, 1989.
9. S. D. Yanowitz and A. M. Bruckstein, "A new method for image segmentation," *Computer Vision, Graphics, and Image Processing*, vol. 46, no. 1, pp. 82–95, Apr. 1989.